# ECON 8803: PS1

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## 1 National Supported Work Demonstration

The data for this problem, `JTRAIN2`, is based on the National Supported Work Demonstration (NSW) experiment.

(a) Run a regression of the outcome (earnings in 1978 in \$1000s) on the training dummy, linearly controlling for age, marriage status, and earnings in 1974. Report the coefficients and standard errors with a reasonable number of significant digits. Provide an interpretation of the coefficient of the treatment dummy $\hat{\beta}$.

| | *1a Regression* |
|---|---|
| | re78 |
| train | 1.758 |
| | (0.633) |
| age | 0.046 |
| | (0.045) |
| married | 0.072 |
| | (0.861) |
| re74 | 0.105 |
| | (0.059) |
| Observations | 445 |
| $R^2$ | 0.028 |
| Adjusted $R^2$ | 0.019 |
| Residual Std. Error | 6.569 (df = 440) |
| F Statistic | $3.131^{**}$ (df = 4; 440) |

- $\hat{\beta}$ here shows us that individuals who participated in the training earned \$1,758 more in 1978 than those who did not receive training

when controlling for age, marital status, and individual earnings in 1974.

(b) Verify that you get the same result for $\hat{\beta}$ if you first run $Y_i$ and the treatment dummy $D_i$ on other covariates, take residuals, and then run the residuals from $Y_i$ regression on the residuals from the $D_i$ regression.

- The coefficient when regressing the residuals from $Y_i$ onto the residuals from $D_i$ is 1.758, equivalent to the initial regression:

| | *1b Regression* |
|---|---|
| | $Y_i$ Residuals |
| $D_i$ Residuals | 1.758 |
| | (0.631) |
| Observations | 445 |
| $R^2$ | 0.017 |
| Adjusted $R^2$ | 0.015 |
| Residual Std. Error | 6.546 (df = 443) |
| F Statistic | $7.760^{***}$ (df = 1; 443) |

(c) Provide an intuitive explanation for why you may get a similar result in (b).

- The results in (a) and (b) are the same because of the Frisch-Waugh-Lovell (FWL) theorem, which states that least squares estimators can be found through a two-step regression using residuals (Hansen).

  After excluding the effects of all control variables in y (residual $y$), exclude the part in training that is a reflection of the person's background (residual $d$), the coefficient is the pure effect of training on wage.

(d) Approximate the CEF of earnings with respect to age.

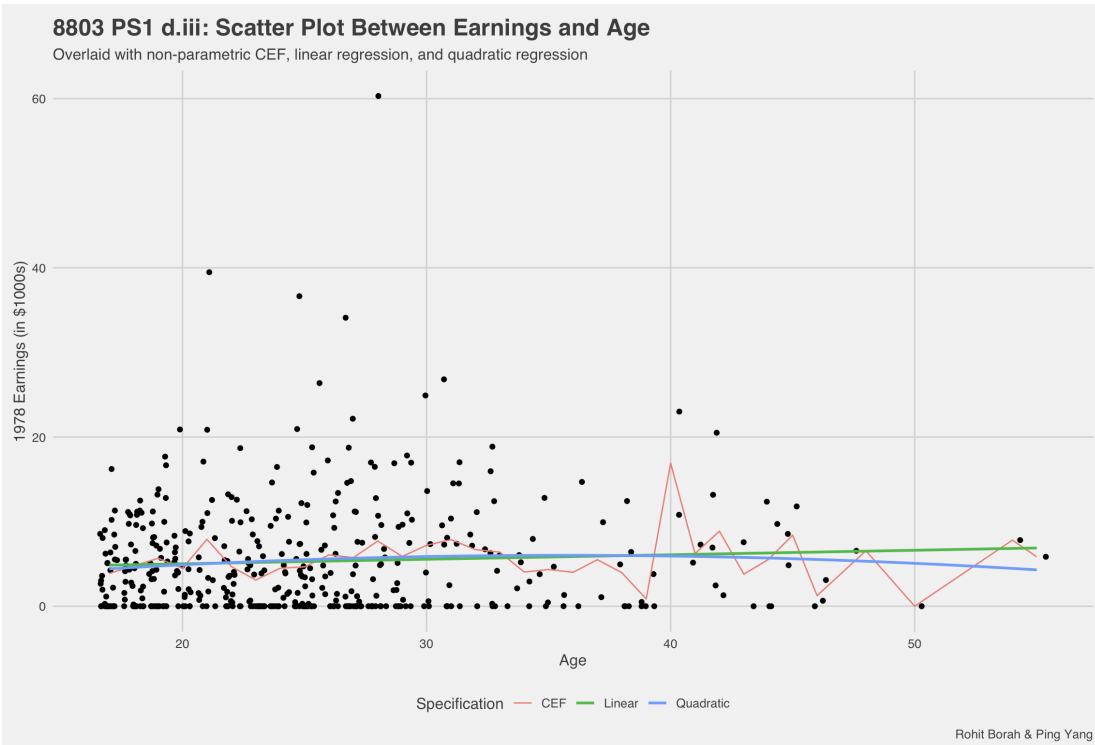
i. Estimate the CEF, $h(.) = \mathbb{E}[\texttt{earnings}_{78}|\texttt{age}]$ non-parametrically.

   - See `df_di` in R.

ii. Approximate the CEF using a linear regression and a quadratic regression of earnings on age.

   - See `lm_dii_linear` and `lm_dii_quad` in R.

iii. Plot the non-parametric CEF, linear, and quadratic fits on the scatter plot between earnings and age. Clearly label which curve corresponds to each specification. What do you observe?




iv. Now, using non-parametric CEF as the proxy of true CEF, calculate the mean square error of linear and quadratic fits. Which specification provides a better approximation and why?

| | |
|---|---|
| MSE: Linear Fit | 0.286 |
| MSE: Quadratic Fit | 0.189 |

- The quadratic fit provides a better approximation than the linear fit as shown by its lower MSE. This shows us that the values from the quadratic fit are closer to the non-parametric CEF than from the linear fit.

(e) Now focus on the estimate of the training dummy $\hat{\beta}$. The estimator can be written as:

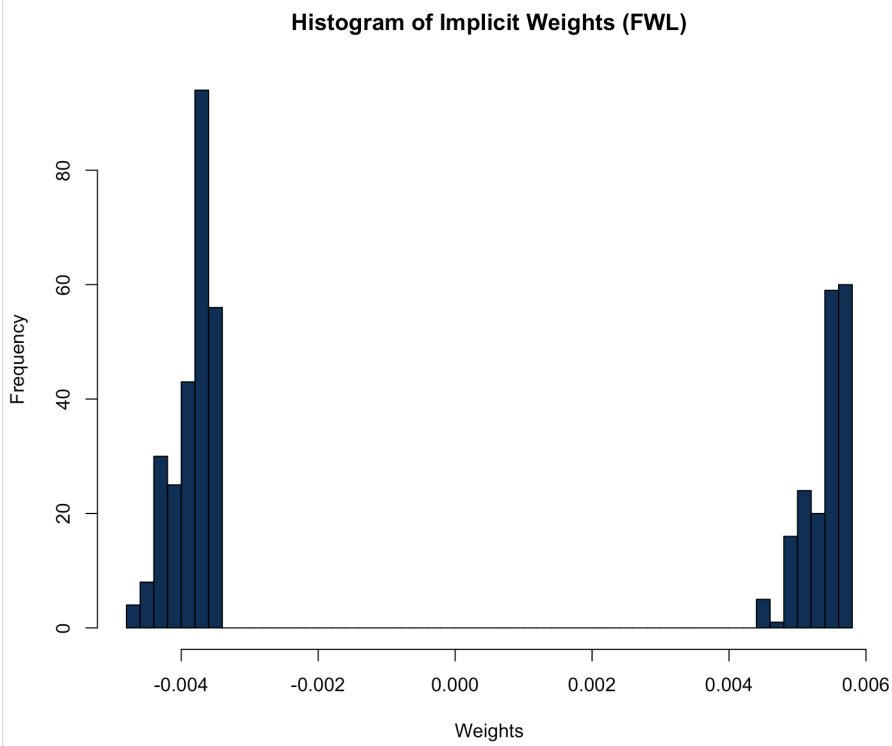
$$\hat{\beta} = \sum_i \omega_i(\mathbf{X}) Y_i$$

where $\omega_i(\mathbf{X})$ are the implicit weights placed on each outcome, which depend on the full regressor matrix $\mathbf{X}$. Recover these weights using the following two techniques:

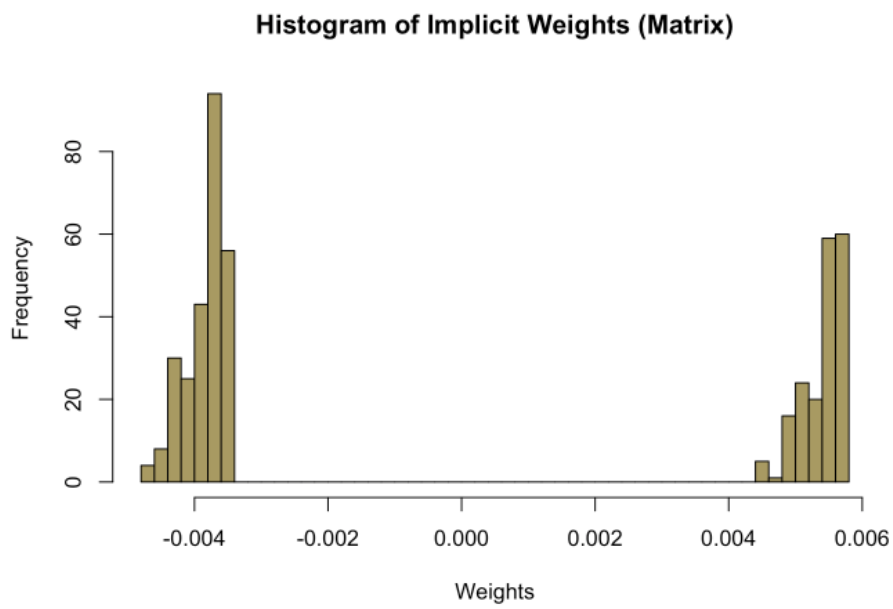
i. Using FWL, derive the expression for these weights:

$$\omega_i = \frac{\bar{D}_i}{\sum_j \bar{D}_j^2}$$

where $\bar{D}_i$ is the residual from regressing the treatment dummy on the controls. Plot the histogram of $\omega_i(\mathbf{X})$ across all observations.



ii. Using the matrix formula $\hat{\beta} = (X'X)^{-1}X'Y$, compute the weights directly. Plot the histogram of $\omega_i(\mathbf{X})$ across all observations. Is it similar to part (i)?




- Yes, it is identical to part (i).

iii. Intuitively, what does the variation in these weights tell you about the identification of the treatment indicator?

- The individuals with high weights are whose treatment status cannot be explained by their demographics. These individuals provide the clean comparison necessary to estimate the treatment effect.

## 2 Nicaragua

For this question, use the `Nicaragua_RCT.csv` file provided on Canvas (Carter 2019). Once the dataset is loaded, you should see the 12 variables listed in the code book. There are important differences between `assign`, `treat`, and `complier`:

**assign** equals 1 if the household was assigned to the early treatment group, i.e. invited to participate in the program. Here, you only see the baseline (round 1) data, so nobody has been treated yet (except for income, where we also observe the round 2 values)

**complier** equals 1 for the households that chose to participate when offered the program

**treat** equals 1 for households that were assigned to the treatment group (`assign` = 1) and who wanted to participate (`complier` = 1) actually received the treatment between the baseline and the midline surveys.

The outcome variable of interest (`income`) is the household's farm income, i.e. the monetized value of the outputs from the specific type of farming that they do. The other baseline variables (`rubro`, `age`, `education`, `mobcap`, `fixcap`, `land`) were all observed before assignment and treatment occurred, and `income_r1` is also the baseline value of income.

(a) Under successful randomization, what would we expect the value of the following expression to be?

$$\mathbb{E}[complier_i | assign_i = 1] - \mathbb{E}[complier_i | assign_i = 0]$$

Compute the sample analogues of the above expression. Use a difference-of-means test to check if the difference is statistically significant. What do you observe?

- We would expect this value to be zero under truly random treatment assignment.
- **Computation**:

$$\begin{aligned} E[complier_i | assign_i = 1] - \mathbb{E}[complier_i | assign_i = 0] \\ = 0.668 - 0.696 \\ = -0.028 \end{aligned}$$

- **Difference-of-means test**: p-value = 0.288, fail to reject $\mathbb{H}_0$
- **Observation**: Although the assignment is not perfectly random as the difference between the two groups is not 0, there is no statistically significant difference between the means of the two groups. We can proceed!

(b) Under successful randomization, what other variable(s) should be uncorrelated with `assign`? Again, comparing sample means among these variables, is there any evidence that treated and control households are unbalanced?

- With successful randomization, we would expect all of the baseline variables to be uncorrelated with `assign` because they were observed prior to treatment assignment.

| Variable | Mean (Treated) | Mean (Control) | Difference | P-value | Significant |
|---|---|---|---|---|---|
| complier | 0.668 | 0.696 | −0.028 | 0.288 | No |
| income_r1 | 7,576.573 | 8,795.342 | −1,218.769 | 0.056 | No |
| income_r2 | 10,250.516 | 10,709.649 | −459.133 | 0.584 | No |
| rubro | 1.728 | 1.824 | −0.096 | 0.011 | Yes |
| age | 52.266 | 54.539 | −2.273 | 0.002 | Yes |
| education | 3.926 | 3.968 | −0.042 | 0.861 | No |
| mobcap | 3,916.307 | 3,800.981 | 115.326 | 0.799 | No |
| fixcap | 3,381.100 | 3,514.176 | −133.076 | 0.740 | No |
| land | 37.201 | 47.551 | −10.350 | 0.017 | Yes |

- There is a statistically significant difference in means for three variables: rubro, age, and land.

(c) We can define the average treatment effect on the treated (ATT), that is the effect of `treat` on `income` among the treated households, using the following linear model:

$$income_i = \alpha + \beta_{ATT} \times treat_i + \epsilon_i$$

If we estimate this model on the full set of households using OLS, will $\hat{\beta}$ be an unbiased estimate of $\beta_{ATT}$? Why or why not?

- No, we do not expect $\hat{\beta}$ to be an unbiased estimate of $\beta_{ATT}$ due to the selection bias - the groups are not randomly assigned. To test, we ran Pearson's product-moment correlation to test the relationship between compliance and baseline income and found a statistically significant correlation (p = 0.008) between the two, indicating pre-treatment bias. This aligns with the descriptive statistics:

$$\mathbb{E}[income_r1|complier = 1] = \text{C\$}8,718$$
$$\mathbb{E}[income_r1|complier = 0] = \text{C\$}6,924$$

(d) Now, estimate two sets of specifications for Equation 2, one without any covariates and one with the baseline variables (`rubro, age, education, mobcap, fixcap, land`). Provide an interpretation of $\hat{\beta}_{ATT}$ for both cases. Do they differ? Why or why not?

- Estimate without covariates:

Table 1:

| | *Dependent variable:* |
|---|---|
| | income_r2 |
| treat | 949.41 |
| | (860.425) |
| Constant | 10,100.49*** |
| | (533.744) |
| Observations | 1,224 |
| $R^2$ | 0.001 |
| Adjusted $R^2$ | 0.0002 |
| Residual Std. Error | 14,646.390 (df = 1222) |
| F Statistic | 1.218 (df = 1; 1222) |
| *Note:* | *p<0.1; **p<0.05; ***p<0.01 |

- Estimate with covariates:

Table 2:

| | *Dependent variable:* |
|---|---|
| | income_r2 |
| treat | 1,262.018* |
| | (736.476) |
| rubro | 3,183.558*** |
| | (562.853) |
| age | 29.283 |
| | (29.315) |
| education | 238.374** |
| | (96.694) |
| mobcap | 0.603*** |
| | (0.055) |
| fixcap | 0.052 |
| | (0.056) |
| land | 42.510*** |
| | (5.202) |
| Constant | −2,400.393 |
| | (1,936.602) |
| Observations | 1,212 |
| $R^2$ | 0.276 |
| Adjusted $R^2$ | 0.271 |
| Residual Std. Error | 12,335.210 (df = 1204) |
| F Statistic | 65.455*** (df = 7; 1204) |

- **Comparison Table**:

| Spec | Treatment Effect | Standard Error | p-value | $r^2$ |
|---|---|---|---|---|
| **Without Covariates** | 949.40 | 860.40 | 0.270 | 0.001 |
| **With Baseline Variables** | 1,262.02 | 736.48 | 0.087 | 0.276 |

- **Interpretation**: In the first specification, the model finds that treated households earn on average C$949.40 more than households that did not participate. However, the $r^2$ value shows that this model does not effectively account for the variation.

  When running the model with controls for the baseline variables, we find that participating households receive C$1,262.02 more.